

Revisiting the Promise of Automated MEP Quantity Takeoff

From IFC Data to Engineering Quantity

IFC



Engineering Meaning



Quantity Takeoff

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ENMA makes openBIM evidence practical

Industry-led collaboration advances practical openBIM research through more than 100 working meetings

RESEARCH EVIDENCE

100+
ENMA-WG Meetings

Continuous practical openBIM research

TECHNICAL ECOSYSTEM

JIZO — IFC validation

Bonsai — native IFC authoring

IFC.js Viewer — model visualization

IfcTester — IDS validation

AI / MCP — knowledge extraction



SUBMITTING PARTY + STAKEHOLDERS

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Continuous, industry-led research transforms openBIM from data exchange into engineering knowledge.

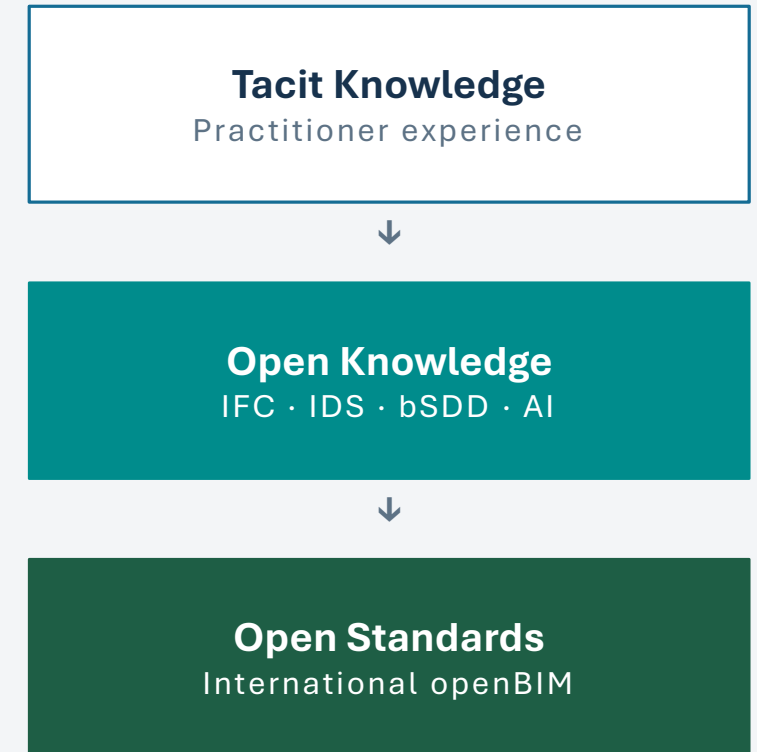


Engineering Knowledge for Everyone

Across the design, construction and operation of building services, practitioners hold a vast body of hard-won engineering knowledge. Most of it stays tacit — locked inside individual and corporate experience — and BIM data alone cannot express it.

ENMA does not simply collect this knowledge. Its mission is to transform tacit engineering knowledge into open engineering knowledge using IFC, IDS, bSDD and AI — and to bridge that knowledge with international openBIM standards so that everyone can use it.

KNOWLEDGE TRANSFORMATION



Engineering Meaning Automation transforms tacit engineering knowledge into open engineering knowledge.

PHILOSOPHY

| Geometry alone is not enough.

| Properties alone are not enough.

| AI alone is not enough.

Engineering Knowledge completes openBIM.

Engineering Meaning Automation begins where geometry and properties alone are no longer sufficient.

Revisiting the Promise

Why automatic MEP quantity takeoff is harder than it looks

THE PROMISE

BIM Model



Automatic Quantity Takeoff

Digital models should make quantities automatically measurable.



THE REALITY

Geometry

Pipe length can be measured.

Properties

Required information may be incomplete.

Engineering Context

Construction conditions affect what must be estimated.

PRACTICAL MEP EXAMPLE

Pipe Length	+	Construction Location	+	Engineering Rules	→	Insulation Quantity
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If the construction location is unknown, the required insulation specification may not be determined.



OUR QUESTION

What is required to turn IFC data into estimable MEP quantities?

MEP quantity takeoff requires engineering meaning, not geometry alone.

Measurement is not the same as estimation.

ENMA Approach to MEP Quantity Takeoff

Automated MEP quantity takeoff moves beyond geometry to engineering meaning and traceable decisions

IFC DATA → ENGINEERING MEANING → ENGINEERING QUANTITY

TRANSFORMATION

IFC Data



Engineering
Meaning



Engineering
Quantity

ROLES IN THE FLOW

IFC Data



IDS Validation



bSDD Semantics



Engineering Knowledge &
Rules



AI-assisted Interpretation



Engineering Quantity

Elements · Properties · Quantities ·
Ports

Check required information

Resolve shared engineering
meaning

Apply engineering context and
estimation logic

Support incomplete or ambiguous
information

Quantity suitable for engineering use

PRACTICAL EXAMPLE

Pipe Length



Construction Location



Engineering Rules



Insulation
Requirement

Illustrative example — not implemented in the current
PoC.

From IFC Data to Engineering Quantity

openBIM provides the information foundation. Engineering knowledge gives that information practical meaning.

Real IFC PoC Results

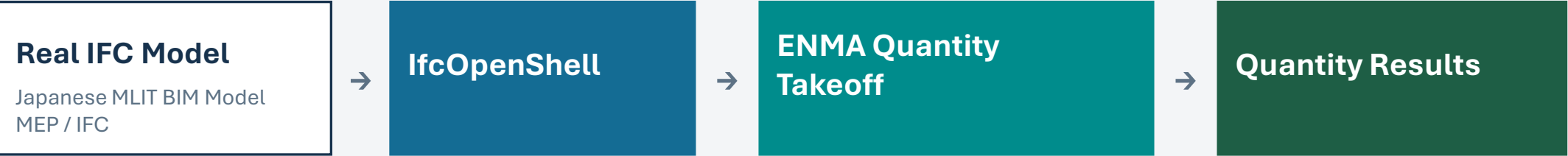
From Real BIM Data to Measurable MEP Quantities

CURRENT PoC SCOPE

Pipes **Implemented**

Ducts **In Progress**

Equipment **Future**



252

Pipe Segments

649.090 m

Total Pipe Length

HORIZONTAL

375.963 m

VERTICAL

273.127 m

SLOPED

0.000 m

81

Pipe Fittings

IFC DATA ANALYZED

Storey

System

Diameter

Length

Direction

Height

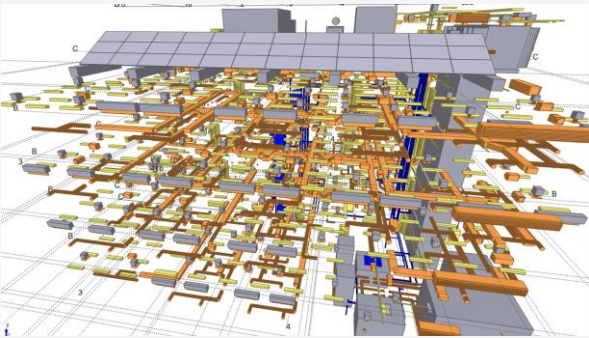
Fittings

Ports

This is not a conceptual example.

These quantities were extracted from an actual IFC model.

Source BIM: Japanese MLIT BIM Model



Actual IFC Data



MEP Elements Identified



Measurable Quantities Produced

What Does “Length” Mean?

From Measured Length to Estimable Quantity

Current Pipe PoC

Centerline-based length

Fitting length not yet deducted

01 Property Length

Length stored in IFC properties / quantities

03 Centerline Length

Length measured along the pipe centerline

02 Geometry Length

Length calculated from IFC geometry

04 Net Pipe Length

Pipe material length considering fittings

05 Quantity for Estimation

Quantity selected after applying engineering rules

Which quantity should be used for estimation?

The measurable quantity is not always the estimable quantity.

Engineering rules are required to determine the quantity used for estimation.

Engineering Meaning



Human Review



Engineering Rules



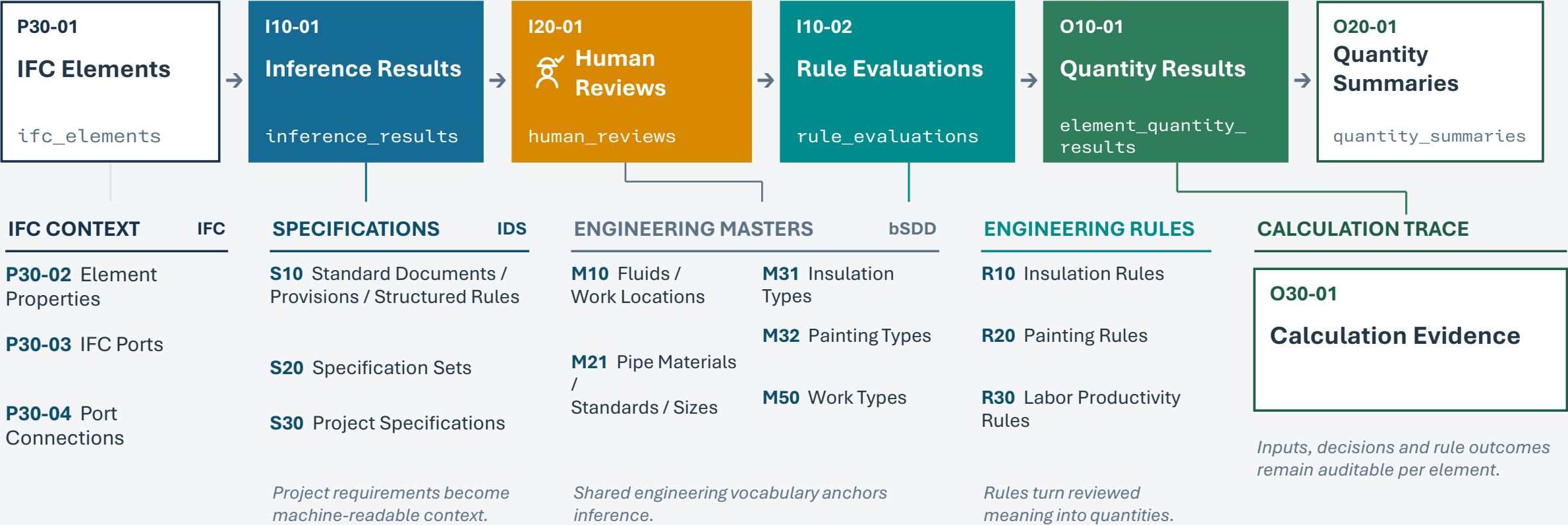
Quantity

ENMA Quantity Takeoff Data Model

From IFC Data to Engineering Quantity

CURRENT PoC SCOPE

Pipes: Implemented · Ducts: In Progress
Equipment: Future



IFC Data → Engineering Meaning → Human Validation → Engineering Rules → Quantity

A traceable data structure: openBIM inputs become engineer-validated, rule-evaluated MEP quantities.

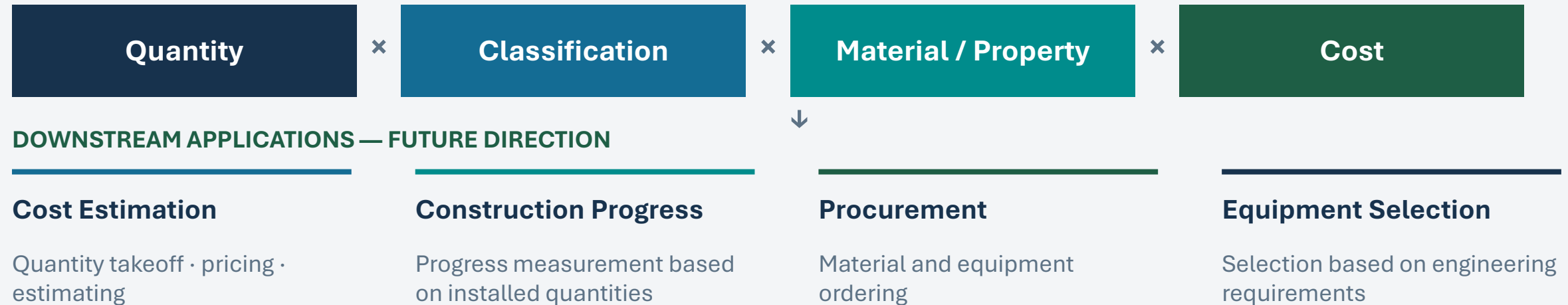
Beyond Quantity Takeoff

Quantity is not the final goal — it is the starting point

CURRENT PoC STATUS

Pipes: Implemented · Ducts: In Progress
Equipment: Future

CONNECTED ENGINEERING INFORMATION



PIPE PoC FUTURE EXTENSION

Pipe Length + Diameter + Material / Standard + Cost Information

↓
Estimation · Procurement

Illustrative direction beyond the current PoC — pipe quantity extraction is implemented; downstream uses are not.

**Quantity takeoff is not the final goal.
It is the starting point for reusable engineering information.**

The same openBIM information can support multiple downstream engineering workflows.

ENGINEERING QUANTITY → ESTIMATION · CONSTRUCTION · PROCUREMENT · EQUIPMENT SELECTION

ENMA 3.0 Vision

Open Engineering Knowledge Platform

CURRENT BIM ENGINEERING WORKFLOW

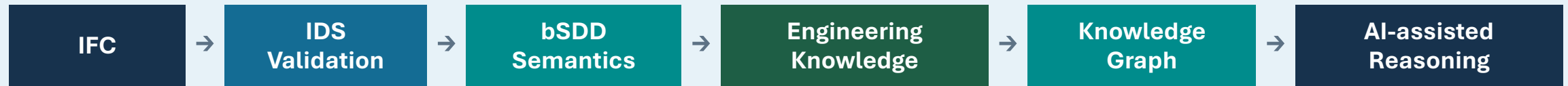
EXISTING TOOLS REMAIN IN PLACE

Architectural BIM → MEP Design → Load Calculation → Equipment Selection → Energy Simulation → Detailed Design → Construction → Facility Management

Existing engineering workflows and tools remain in place.

OPEN ENGINEERING KNOWLEDGE PLATFORM

VENDOR-NEUTRAL KNOWLEDGE LAYER BASED ON openBIM STANDARDS



Engineering knowledge and human engineering judgment remain central.

Interoperable data → validated requirements → shared semantics → reusable rules and intent

ENGINEERING APPLICATIONS

ONE FOUNDATION • MULTIPLE APPLICATIONS



PoC DEMONSTRATED

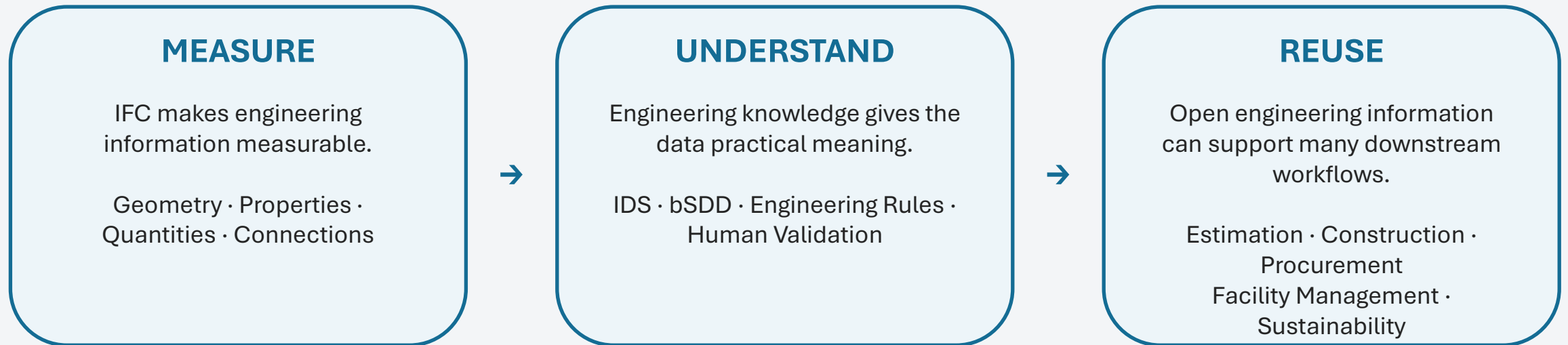
Quantity Takeoff is one application of reusable open engineering knowledge.

ENMA enriches existing BIM workflows with open standards, engineering knowledge and AI-assisted reasoning.

openBIM connects data. Engineering Knowledge connects decisions.

From Quantity Takeoff to Open Engineering

What we learned from revisiting automated MEP quantity takeoff



Quantity takeoff is not the final goal. It is one application of openBIM data.

The real opportunity is to transform BIM data into reusable engineering knowledge.

Open standards allow engineering information to continue beyond the original authoring software.

ENMA is still at an early stage. Our Quantity Takeoff PoC is one practical step toward reusable open engineering knowledge.

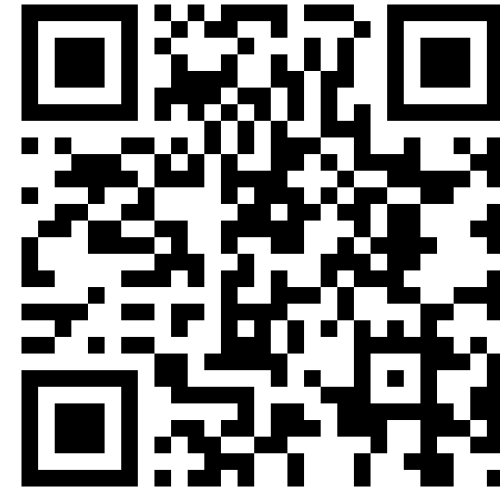
Engineering Knowledge for Everyone

Thank You

Engineering Knowledge for Everyone

Let's build Open Engineering together.

openBIM · IFC · IDS · bSDD · Engineering Knowledge · AI-assisted Interpretation



ENMA-WG GitHub

<https://github.com/ENMA-WG/enma-poc>

Explore the PoC · Review the approach
Share your feedback

#bSSTokyo26